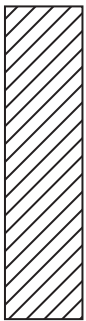


7 The components of a cell membrane determine its properties.

The membrane is a phospholipid bilayer.

(a) The table shows one way that the components of phospholipids can be drawn.

Component	Way of drawing the component
phosphate group	
glycerol	
fatty acid	
bond	

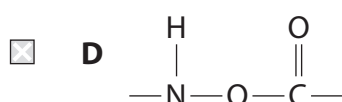
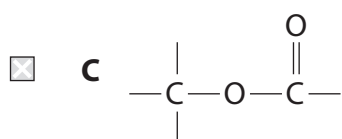
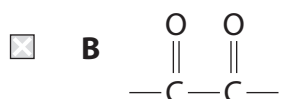
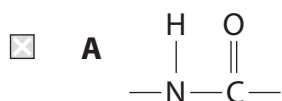
(i) Draw a diagram of a phospholipid, using the information in the table.

(2)

(ii) One of the bonds present in phospholipids is an ester bond.

Which diagram shows an ester bond?

(1)



(b) The fluidity of a membrane is increased by a number of factors:

- increase in temperature
- increase in the proportion of unsaturated fatty acids
- decrease in cholesterol
- fatty acids with shorter side chains.

(i) Explain why an increase in temperature increases the fluidity of the membrane.

(2)

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- (ii) Which row in the table describes an unsaturated fatty acid compared with a saturated fatty acid with the same number of carbon atoms?

(1)

	Has double bonds between two carbons	Number of hydrogens
☐ A	no	fewer than the saturated fatty acid
☐ B	no	more than the saturated fatty acid
☐ C	yes	fewer than the saturated fatty acid
☐ D	yes	more than the saturated fatty acid

- (iii) Explain why a decrease in cholesterol increases the fluidity of the membrane.

(2)

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- (iv) Suggest why a fatty acid with a shorter side chain will increase the fluidity of a membrane.

(1)

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(c) Studies have shown that when fish are moved into water at a lower temperature, the fatty acid content of their membranes changes.

- (i) Suggest how the content of the fatty acids in these membranes may have changed.

(2)

- (ii) Suggest why these changes are necessary for the survival of the fish.

(2)

(Total for Question 7 = 13 marks)

